The medieval cosmos so marvellously sung by Dante was interpreted on the basis of a hierarchical organization of the universe which reflected the hierarchical organization of European society: a spherical cosmic structure with the Earth at its centre; the irreducible separation between Earth and heavens; finalistic and metaphorical explanations of natural phenomena. Fear of God, fear of death; little attention to nature; the idea that forms preceding things determine the structure of the world; the idea that the source of knowledge could only be the past, in revelation and tradition …

There is none of this in the world of Democritus as sung by Lucretius. There is no fear of the gods; no ends or purposes in the world; no cosmic hierarchy; no distinction between Earth and heavens. There is a deep love of nature, a serene immersion within it; a recognition that we are profoundly part of it; that men, women, animals, plants and clouds are organic threads of a marvellous whole, without hierarchies. There is a feeling of deep universalism in the wake of the splendid words of Democritus: 'To a wise man, the whole earth is open, because the true country of a virtuous soul is the entire universe.'[25]

There is, too, the ambition of being able to think about the world in simple terms. Of being able to investigate and understand the secrets of nature. To know more than our parents. And there are extraordinary conceptual tools on which Galileo, Kepler and Newton will build: the idea of free rectilinear motion in space; the idea of elementary bodies and their interactions, out of which the world is constructed; the idea of space as a container of the world.

And there is the simple idea of the finite divisibility of things. The granular quality of the world. The idea which stops the infinite between our fingers. This idea is at the root of the atomic hypothesis, but it will also return with augmented force with quantum mechanics, and today is revealing itself to be powerful again – as the keystone of quantum gravity.

The first person to make the parts of the mosaic which begin to emerge from Renaissance naturalism cohere – and to reprise the Democritean vision, immensely reinforced, placing it at the centre of modern thought – will be an Englishman, the greatest scientist of all time and the first protagonist of the following chapter.